QUALITY INFORMATION DOCUMENT

For OMI_CLIMATE_OHC_MEDSEA

Issue: 1.4

Contributors : Vladyslav Lyubartsev, Pietro Miraglio, Ali Aydogdu, Lucia Gualtieri, Emanuela Clementi, Simona Masina, Nadia Pinardi

Approval Date: November 2025

MERCATOR OCEAN
INTERNATIONAL
2 avenue de l'Aérodrome de Montaudran, 31400 Toulouse, FRANCE
Tél : +33 5 61 39 38 02 - Fax : +33 5 61 39 38 99
Société civile de droit français au capital de 2 000 000 € - 522 911 577 RCS Toulouse - SIRET 522 911 577 00024

marine.copernicus.eu
mercator-ocean.eu

RECORD TABLE

When the quality of the products changes, the QuID is updated and a row is added to this table. The third column specifies which sections or sub-sections have been updated. The fourth column should mention the version of the product to which the change applies.

Issue	Date	§	Description of Change	Author	Validated By
1.0	2018/09/10	All	Creation of the document	Emanuela Clementi, Vladyslav Lyubartsev	Emanuela Clementi
1.1	2018/09/10	All	Updated document after MO reviews	Emanuela Clementi, Vladyslav Lyubartsev	Emanuela Clementi
1.2	2021/01/05	All	Updated document using new MED Reanalysis	Emanuela Clementi, Vladyslav Lyubartsev	Emanuela Clementi
1.3	2022/06/16	All	Extension to 2019	Emanuela Clementi	Anna Chiara Goglio
1.4	2025/05/28	All	Extension of the time series, Setup of operational temporal extension and renaming of the product name	Lucia Gualtieri	Emanuela Clementi

TABLE OF CONTENTS

RECORD TABLE **2**

1) INTRODUCTION **4**

a) Product Name 4

b) Indicator Evaluation Method 4

c) Copernicus Marine products used as input for the computation 5

2) SCIENTIFIC VALIDATION **6**

a) Mediterranean Ocean Heat Content Anomaly (0-700m) 6

3) REFERENCES **7**

1) INTRODUCTION

a) Product Name

OMI_CLIMATE_OHC_MEDSEA_area_averaged_anomalies

b) Indicator Evaluation Method

Ocean heat content is defined here as the deviation from a reference period (1993-2014) and is closely proportional to the average temperature change from z_1 = 0 m to z_2 = 700 m depth:

$$\mathrm{OHC} = \int_{z_1}^{z_2} \rho_0 \, c_p \left(T_{yr} - T_{clim}\right) dz \qquad [1]$$

with a reference density of = 1030 kgm^{-3} and a specific heat capacity of c_p = 3980 J kg^{-1} °C^{-1} (e.g. von Schuckmann et al., 2009).
Time series of annual mean values area averaged ocean heat content is provided for the Mediterranean Sea (30°N, 46°N; 6°W, 36°E) and is evaluated in areas where the topography is deeper than 700m.

c) Copernicus Marine products used as input for the computation

Product name (and Type)	Documentation
MEDSEA_MULTIYEAR_PHY_006_004	QUID: https://catalogue.marine.copernicus.eu/documents/QUID/CMEMS-MED-QUID-006-004.pdf PUM: https://catalogue.marine.copernicus.eu/documents/PUM/CMEMS-MED-PUM-006-004.pdf
GLOBAL_REANALYSIS_PHY_001_031	QUID : https://catalogue.marine.copernicus.eu/documents/QUID/CMEMS-GLO-QUID-001-031.pdfPUM : https://catalogue.marine.copernicus.eu/documents/PUM/CMEMS-GLO-PUM-001-031.pdf
INSITU_GLO_TS_REP_OBSERVATIONS_013_001_B	QUID : https://catalogue.marine.copernicus.eu/documents/QUID/CMEMS-INS-QUID-013-001b.pdf PUM : https://catalogue.marine.copernicus.eu/documents/PUM/CMEMS-INS-PUM-013-001-b.pdf
MULTIOBS_GLO_PHY_TSUV_3D_MYNRT_015_012	QUID : https://catalogue.marine.copernicus.eu/documents/QUID/CMEMS-MOB-QUID-015-012.pdf PUM : https://catalogue.marine.copernicus.eu/documents/PUM/CMEMS-MOB-PUM-015-012.pdf

2) SCIENTIFIC VALIDATION

a) Mediterranean Ocean Heat Content Anomaly (0-700m)

More details and full scientific evaluation can be found in the Copernicus Marine Ocean State Report issue 2 (OSR2) Section 2.1 (von Schuckmann et al., 2018). With respect to the results shown in OSR2, an updated higher resolution Mediterranean Sea Reanalysis has been used to evaluate the present OMI.
The quality evaluation of OMI_CLIMATE_OHC_MEDSEA_area_averaged_anomalies is based on the "multi-product" approach as introduced in the second issue of the Ocean State Report (von Schuckmann et al., 2018a) following the MyOcean's experience (Masina et al., 2017). Six global products and a regional (Mediterranean Sea) product have been used to build an ensemble mean and its associated ensemble spread (Figure 1, dark red line and shaded area). For each dataset, all available data is used thus resulting in a different time extension. The Mediterranean Reanalysis and Interim timeseries are used covering up to the year before the current one [-1Y], and it is operationally extended on a yearly basis. The ensemble mean is calculated over the time period common to all datasets.
Results of Fig. 1 indicate how each member is distributed around the ensemble mean and show a continuous increase over the period 1993-2021 at rate of 1.38±0.08 W/m2 in the upper 700m. After 2005 the rate has clearly increased with respect the previous decade, in agreement with Iona et al. (2018).

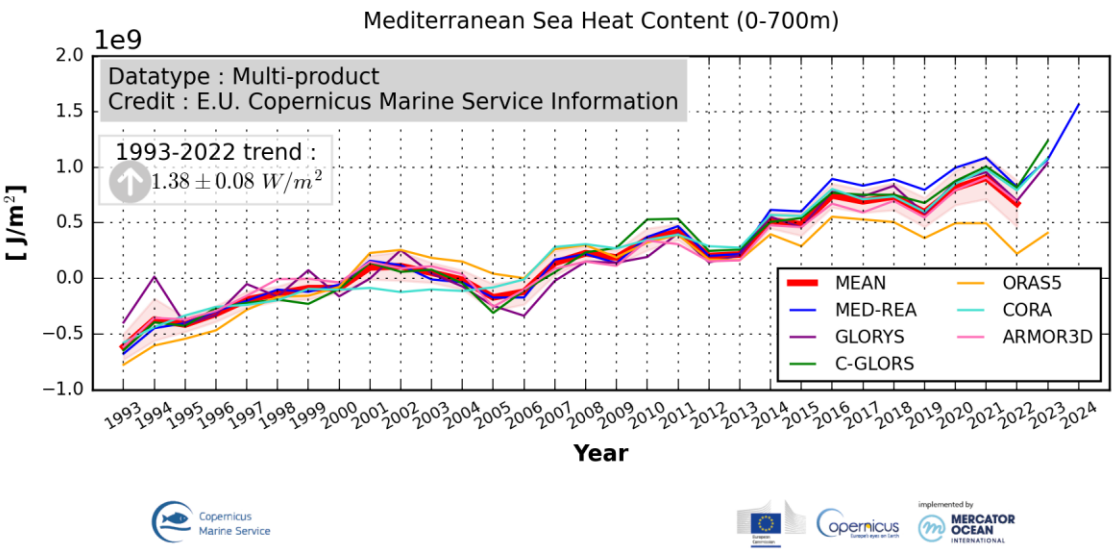

Figure 1: *Time series of area averaged ocean heat content in the Mediterranean Sea, and integrated over the 0-700m depth layer. Time series are based on different data products, i.e. Mediterranean Sea Reanalysis (dark blue), global ocean reanalysis GLORYS (purple), C-GLORS (green), ORAS5 (orange), FOAM (dark green); global observational based products CORA (light blue), ARMOR3D (pink), and the ensemble mean (dark red) and the ensemble spread (red shaded). Details on the products are given in the corresponding PUM for this OMI.*

3) REFERENCES

Escudier, R., Clementi, E., Cipollone, A., Pistoia, J., Drudi, M., Grandi, A., Lyubartsev, V., Lecci, R., Aydogdu, A., Delrosso, D., Omar, M., Masina, S., Coppini, G., Pinardi, N. 2021. A High Resolution Reanalysis for the Mediterranean Sea. Frontiers in Earth Science, Vol.9, pp.1060, DOI:10.3389/feart.2021.702285.

Iona, A., A. Theodorou, S. Sofianos, S. Watelet, C. Troupin, J.-M. Beckers, 2018: Mediterranean Sea climatic indices: monitoring long term variability and climate changes, Earth Syst. Sci. Data Discuss., https://doi.org/10.5194/essd-2018-51, in review.

Masina S., A. Storto, N. Ferry, M. Valdivieso, K. Haines, M. Balmaseda, H. Zuo, M. Drevillon, L. Parent, 2017: An ensemble of eddy-permitting global ocean reanalyses from the MyOcean project. Climate Dynamics, 49 (3): 813-841. DOI: 10.1007/s00382-015-2728-5

von Schuckmann, K., F. Gaillard and P.-Y. Le Traon, 2009: Global hydrographic variability patterns during 2003-2008, Journal of Geophysical Research, 114, C09007, doi:10.1029/2008JC005237.

von Schuckmann et al., 2018: Ocean heat content. In: The Copernicus Marine Environment Monitoring Service Ocean State Report, issue 2, Journal of Operational Oceanography, 11:sup1, s1-s142, DOI: 10.1080/1755876X.2018.1489208.